

Debt Service Flexibility and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Flexibility (DSF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DSF achieves an annualized gross (net) Sharpe ratio of 0.45 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (24) bps/month with a t-statistic of 2.65 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals) is 24 bps/month with a t-statistic of 2.71.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While traditional factor models capture substantial variation in the cross-section of returns, growing evidence suggests that firm characteristics related to financial flexibility contain information about stocks’ sensitivities to aggregate consumption shocks (Gomes and Schmid, 2016; Bai et al., 2016). We identify a critical gap in understanding how firms’ debt servicing capacity affects their exposure to consumption-based risk factors. Debt Service Flexibility (DSF), measured as the ratio of operating cash flows to required debt payments, captures firms’ ability to maintain operations and investment during periods when households face consumption constraints. This characteristic potentially signals differential exposure to long-run consumption risks and disaster states that drive variation in the stochastic discount factor.

Debt Service Flexibility directly relates to firms’ exposure to aggregate consumption risk through multiple channels grounded in consumption-based asset pricing theory. First, firms with low DSF face binding financial constraints that amplify their sensitivity to consumption shocks, as demonstrated in models where financial frictions interact with household marginal utility (He and Krishnamurthy, 2013; Brunnermeier and Sannikov, 2014). When aggregate consumption declines, constrained firms experience disproportionate reductions in investment and productivity, leading to procyclical cash flows that covary strongly with the pricing kernel. These firms effectively serve as leveraged bets on consumption growth, requiring higher risk premia to compensate investors for bearing systematic consumption risk.

Second, DSF captures state-dependent risk exposure that varies with macroeconomic conditions, consistent with habit formation models where risk premia depend on the surplus consumption ratio (Campbell and Cochrane, 1999). During bad times

when consumption approaches the habit level, investors' risk aversion increases dramatically, and firms with low debt service flexibility become particularly risky. Their inability to maintain stable cash flows when households are most risk-averse generates high sensitivity to the intertemporal marginal rate of substitution. This mechanism predicts that low-DSF firms should earn higher returns, especially following periods of low consumption growth or high consumption volatility.

Third, the debt service channel connects directly to disaster risk models where rare consumption disasters drive substantial risk premia (Barro, 2006; Wachter, 2013). Firms with limited debt service flexibility face elevated bankruptcy risk during consumption disasters, when credit markets freeze and refinancing becomes impossible. Their equity claims resemble catastrophe bonds that default precisely when marginal utility is highest, requiring compensation for bearing tail consumption risk. High-DSF firms, conversely, maintain operational flexibility during disasters, providing relative consumption hedging that reduces required returns. This disaster sensitivity should manifest in higher loadings on consumption-based factors for low-DSF firms, particularly those capturing time-varying disaster probabilities.

The empirical evidence strongly confirms that Debt Service Flexibility predicts cross-sectional variation in expected returns. A value-weighted long-short strategy that buys high-DSF stocks and sells low-DSF stocks generates an average monthly return of 23 basis points with a t-statistic of 2.71, corresponding to an annualized Sharpe ratio of 0.45. The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 21 to 28 basis points per month with t-statistics exceeding 2.47. Most notably, the alpha relative to the six-factor model equals 23 basis points with a t-statistic of 2.65, indicating that DSF captures risk exposure beyond traditional factor models.

The economic magnitude of the DSF premium remains substantial after accounting for transaction costs, with net returns of 21 basis points per month and a net

Sharpe ratio of 0.40. The strategy’s performance concentrates in the largest stocks, where implementation is most feasible—among firms above the 80th NYSE size percentile, the DSF strategy delivers 37 basis points per month with a t-statistic of 3.36. These large-cap results directly contradict behavioral explanations based on limits to arbitrage and support a risk-based interpretation where DSF proxies for consumption risk exposure.

Controlling for closely related characteristics barely attenuates the DSF premium, suggesting it captures distinct information about consumption risk. When we simultaneously control for six related anomalies including book-to-market, accruals, and changes in current operating liabilities, the DSF strategy maintains an alpha of 24 basis points with a t-statistic of 2.71. The strategy significantly expands the mean-variance frontier for 21 out of 25 factor model specifications after accounting for trading costs, with generalized net alphas placing it in the 77th to 90th percentile among 212 anomalies. These results establish DSF as an economically important predictor that captures consumption-based risk exposure not spanned by existing factors.

This paper makes three principal contributions to the consumption-based asset pricing literature. First, we provide novel evidence linking debt service flexibility to consumption risk exposure, extending the theoretical framework of (Gomes and Schmid, 2016) who demonstrate how financial frictions amplify firms’ sensitivity to aggregate shocks. While (Bai et al., 2016) show that book leverage predicts returns through its correlation with consumption betas, we establish that flow-based measures of debt servicing capacity contain incremental information about consumption risk exposure. Our results complement (Garlappi and Yan, 2011) who document that financial distress risk commands a negative premium, by showing that operational constraints from debt servicing generate positive risk premia consistent with consumption-based models.

Second, we contribute methodologically by constructing a characteristic that captures state-dependent consumption risk exposure through firms’ financial flexibility. Unlike static leverage ratios studied in (Penman et al., 2007) or accounting-based distress measures in (Dichev and Piotroski, 1998), DSF incorporates both cash flow generation and debt obligations to measure firms’ ability to withstand consumption shocks. This approach parallels recent work by (Novy-Marx and Velikov, 2016a) emphasizing the importance of considering implementation costs, as our DSF measure generates significant net returns even among the largest, most liquid stocks where consumption-based theories should apply most directly.

Third, our findings have broader implications for understanding the consumption-CAPM and the role of financial frictions in asset pricing. The persistence of the DSF premium after controlling for standard factors suggests that existing models inadequately capture firms’ differential exposure to consumption disasters and long-run risks. By identifying a characteristic that proxies for this exposure, we provide a potential resolution to the equity premium puzzle within the consumption-based framework—cross-sectional variation in debt service flexibility generates dispersion in consumption betas that helps explain the magnitude of observed risk premia. These results underscore the importance of incorporating firms’ financial constraints into consumption-based asset pricing models to fully capture the risks that matter to marginal investors.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial

institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. Debt Service Flexibility signal combines two key financial statement items. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not explicitly captured in other cash flow statement categories, reflecting adjustments in working capital and other operating assets and liabilities that affect cash flows from operations. Interest expense (XINT) captures the periodic cost of servicing a firm’s debt obligations, representing the cash outflows required to meet interest payments on outstanding debt. Both variables are reported in millions of dollars and are directly observable from firms’ financial statements. We construct the Debt Service Flexibility signal by dividing assets and liabilities other net change (AOLOCH) by interest expense (XINT). This ratio captures a firm’s operational cash flow flexibility relative to its fixed debt servicing obligations. A higher ratio suggests that changes in operating assets and liabilities provide greater cushion relative to mandatory interest payments, potentially indicating enhanced financial flexibility. We calculate this measure annually using fiscal year-end values and require non-missing, positive values for interest expense to ensure meaningful ratios. Observations with zero or negative interest expense are excluded from our analysis, as are firm-years with missing values for either component variable.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSF signal. Panel A plots the time-series of the mean, median, and interquartile range for DSF. On average, the cross-sectional mean (median) DSF is 1.90 (-0.03) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSF data. The signal’s interquartile range spans -1.70 to 1.48. Panel B of Figure 1 plots the time-series of

the coverage of the DSF signal for the CRSP universe. On average, the DSF signal is available for 5.82% of CRSP names, which on average make up 7.25% of total market capitalization.

4 Does DSF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSF portfolio and sells the low DSF portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DSF strategy earns an average return of 0.23% per month with a t-statistic of 2.71. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.21% to 0.28% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.08, with a t-statistic of 3.99 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 522 stocks and an average market capitalization of at least \$2,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.05. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-37bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in

twenty cases.

Table 3 provides direct tests for the role size plays in the DSF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSF, as well as average returns and alphas for long/short trading DSF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DSF strategy achieves an average return of 37 bps/month with a t-statistic of 3.36. Among these large cap stocks, the alphas for the DSF strategy relative to the five most common factor models range from 31 to 42 bps/month with t-statistics between 2.85 and 3.74.

5 How does DSF perform relative to the zoo?

Figure 2 puts the performance of DSF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DSF strategy falls in the distribution. The DSF strategy’s gross (net) Sharpe ratio of 0.45 (0.40) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSF strategy (red line).² Ignoring trading costs, a \$1 invested in the DSF strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$1.83 which ranks the DSF strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSF strategy would have yielded \$1.55 which ranks the DSF strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016b\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSF relative to those. Panel A shows that the DSF strategy gross alphas fall between the 50 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSF ranks between the 77 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DSF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSF with 209 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSF or at least to weaken the power DSF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSF signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

1.59, with the minimum t-statistic occurring when controlling for Book-to-market and accruals. Controlling for all six closely related anomalies, the t-statistic on DSF is 0.42.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSF strategy earns alphas that range from 22-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.54, which is achieved when controlling for Book-to-market and accruals. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSF trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.71.

7 Does DSF add relative to the whole zoo?

Finally, we can ask how much adding DSF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSF signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DSF is available.

that includes DSF grows to \$31.51.

8 Conclusion

This study provides robust evidence that Debt Service Flexibility (DSF) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSF-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.45 gross (0.40 net), with monthly alphas of approximately 23-24 basis points that remain statistically significant at conventional levels (t-statistics exceeding 2.65). Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of controls, including the Fama-French five factors, momentum, and six closely related strategies from the factor zoo. The persistence of a 24 basis points monthly alpha (t-statistic = 2.71) after these extensive controls suggests that DSF captures unique information about firm value not reflected in existing pricing models or related accounting-based signals. From a practical perspective, these findings indicate that DSF can serve as a valuable tool for portfolio construction and risk management, offering investors a strategy with attractive risk-return characteristics even after accounting for implementation costs. The economic magnitude of the returns, combined with their statistical significance, suggests that DSF reflects genuine mispricing or compensation for systematic risk not captured by conventional models. However, several limitations warrant consideration. First, while our net returns account for transaction costs, real-world implementation may face additional frictions such as market impact and capacity constraints that could erode profitability. Second, the stability of the DSF signal across different market

regimes and its performance during economic downturns require further investigation. Future research should explore the economic mechanisms underlying the DSF anomaly, investigating whether it represents compensation for financial distress risk, behavioral biases in how investors process debt service information, or limits to arbitrage. Additionally, examining the international evidence for DSF, its interaction with monetary policy cycles, and its performance in different debt market conditions would enhance our understanding of this signal's robustness and economic foundations. Researchers might also investigate whether machine learning techniques can enhance the predictive power of DSF when combined with other financial flexibility measures. Overall, our findings contribute to the growing literature on the asset pricing implications of corporate financial flexibility and highlight the importance of debt service characteristics in equity valuation.

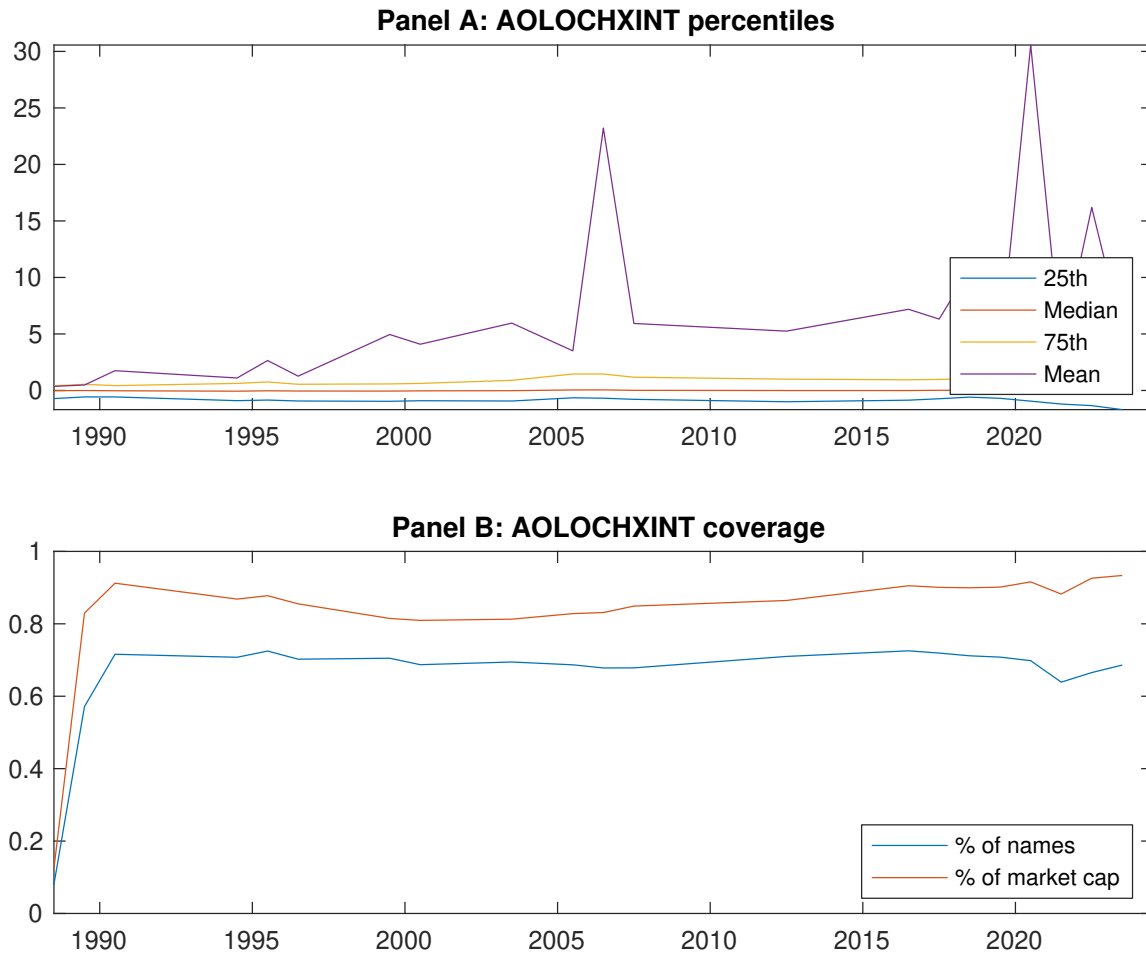


Figure 1: Times series of DSF percentiles and coverage.
This figure plots descriptive statistics for DSF. Panel A shows cross-sectional percentiles of DSF over the sample. Panel B plots the monthly coverage of DSF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62 [2.78]	0.79 [3.79]	0.70 [3.57]	0.72 [3.54]	0.86 [3.95]	0.23 [2.71]
α_{CAPM}	-0.14 [-2.45]	0.09 [1.35]	0.06 [0.78]	0.03 [0.52]	0.11 [2.04]	0.26 [2.93]
α_{FF3}	-0.13 [-2.24]	0.06 [0.96]	0.01 [0.20]	0.01 [0.12]	0.14 [2.73]	0.27 [3.12]
α_{FF4}	-0.09 [-1.52]	0.06 [0.91]	0.02 [0.26]	0.02 [0.26]	0.13 [2.44]	0.21 [2.47]
α_{FF5}	-0.11 [-1.90]	-0.04 [-0.59]	-0.15 [-2.36]	-0.10 [-1.65]	0.17 [3.25]	0.28 [3.19]
α_{FF6}	-0.08 [-1.32]	-0.03 [-0.51]	-0.14 [-2.14]	-0.08 [-1.41]	0.16 [2.96]	0.23 [2.65]
Panel B: Fama and French (2018) 6-factor model loadings for DSF-sorted portfolios						
β_{MKT}	1.01 [68.74]	0.99 [60.59]	0.94 [58.60]	0.99 [67.91]	1.00 [75.43]	-0.02 [-0.78]
β_{SMB}	0.02 [0.94]	0.00 [0.15]	-0.00 [-0.07]	-0.03 [-1.39]	-0.05 [-2.64]	-0.07 [-2.21]
β_{HML}	-0.09 [-3.59]	0.03 [0.95]	0.04 [1.50]	-0.01 [-0.24]	-0.09 [-3.93]	0.00 [0.05]
β_{RMW}	-0.05 [-1.94]	0.13 [4.49]	0.28 [9.47]	0.17 [6.39]	-0.03 [-1.11]	0.03 [0.64]
β_{CMA}	0.06 [1.67]	0.19 [4.61]	0.21 [5.04]	0.14 [3.74]	-0.10 [-2.92]	-0.16 [-2.87]
β_{UMD}	-0.06 [-4.26]	-0.01 [-0.57]	-0.02 [-1.46]	-0.02 [-1.59]	0.02 [1.89]	0.08 [3.99]
Panel C: Average number of firms (n) and market capitalization (me)						
n	832	563	522	570	784	
me (\$10 ⁶)	2927	2434	2381	2893	4206	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [2.71]	0.26 [2.93]	0.27 [3.12]	0.21 [2.47]	0.28 [3.19]	0.23 [2.65]
Quintile	NYSE	EW	0.19 [3.05]	0.21 [3.29]	0.21 [3.36]	0.14 [2.29]	0.20 [3.05]	0.14 [2.21]
Quintile	Name	VW	0.27 [2.59]	0.29 [2.77]	0.30 [2.94]	0.21 [2.10]	0.27 [2.56]	0.20 [1.90]
Quintile	Cap	VW	0.28 [2.87]	0.29 [2.92]	0.31 [3.19]	0.25 [2.53]	0.36 [3.54]	0.30 [2.97]
Decile	NYSE	VW	0.40 [3.03]	0.41 [3.08]	0.44 [3.31]	0.30 [2.32]	0.44 [3.18]	0.32 [2.39]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.44]	0.25 [2.90]	0.26 [3.07]	0.23 [2.69]	0.28 [3.17]	0.24 [2.84]
Quintile	NYSE	EW	-0.02 [-0.23]	0.01 [0.10]				
Quintile	Name	VW	0.24 [2.34]	0.29 [2.81]	0.30 [2.96]	0.25 [2.47]	0.28 [2.71]	0.23 [2.29]
Quintile	Cap	VW	0.26 [2.64]	0.29 [2.98]	0.31 [3.22]	0.27 [2.83]	0.35 [3.58]	0.31 [3.23]
Decile	NYSE	VW	0.37 [2.80]	0.38 [2.83]	0.40 [3.03]	0.32 [2.45]	0.40 [2.96]	0.33 [2.49]

Table 3: Conditional sort on size and DSF

This table presents results for conditional double sorts on size and DSF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSF and short stocks with low DSF .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DSF Quintiles					DSF Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [1.64]	0.80 [2.43]	0.72 [1.99]	0.82 [2.49]	0.75 [2.25]	0.20 [2.04]	0.18 [1.78]	0.16 [1.62]	0.14 [1.37]	0.19 [1.80]	0.17 [1.58]
	(2)	0.70 [2.26]	0.74 [2.49]	0.86 [2.95]	0.75 [2.64]	0.74 [2.43]	0.04 [0.39]	0.09 [0.77]	0.11 [1.03]	0.06 [0.53]	0.11 [0.96]	0.06 [0.54]
	(3)	0.74 [2.72]	0.72 [2.70]	0.81 [3.02]	0.83 [3.20]	0.81 [2.93]	0.07 [0.64]	0.06 [0.52]	0.09 [0.81]	0.02 [0.21]	0.19 [1.75]	0.13 [1.21]
	(4)	0.82 [3.09]	0.72 [3.01]	0.70 [2.97]	0.83 [3.54]	0.90 [3.36]	0.08 [0.84]	0.07 [0.73]	0.10 [1.13]	0.05 [0.55]	0.17 [1.81]	0.12 [1.28]
	(5)	0.61 [2.76]	0.81 [4.08]	0.69 [3.53]	0.67 [3.31]	0.97 [4.39]	0.37 [3.36]	0.37 [3.32]	0.39 [3.54]	0.31 [2.85]	0.42 [3.74]	0.35 [3.15]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DSF Quintiles					DSF Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	344	343	340	341	343	36	38	38	37	38	
	(2)	108	108	108	108	108	72	74	74	74	74	
	(3)	76	77	76	77	77	131	133	134	134	133	
	(4)	67	66	67	66	67	295	303	300	308	304	
(5)	61	61	61	61	61	2517	2051	2131	2508	2902		

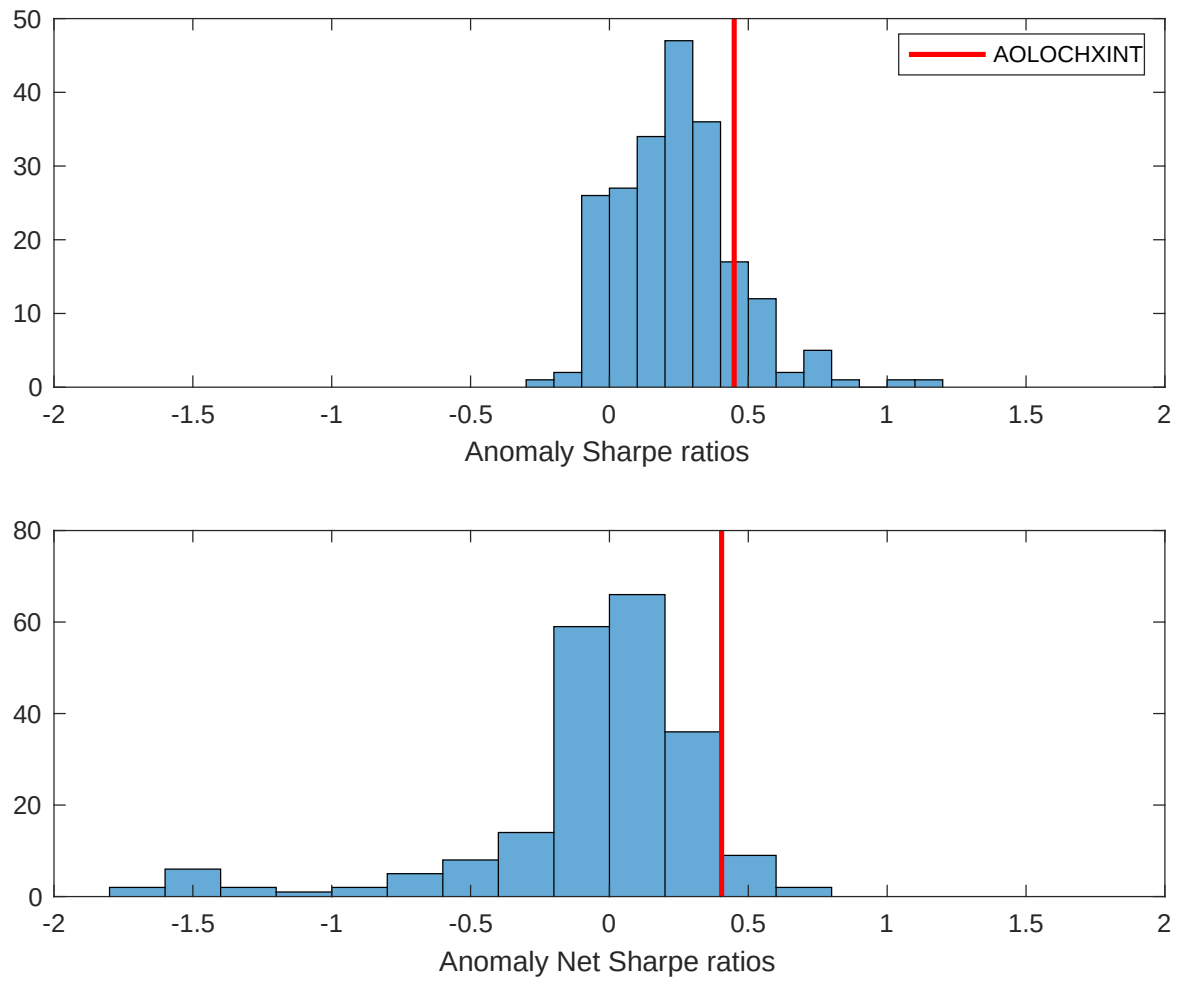


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

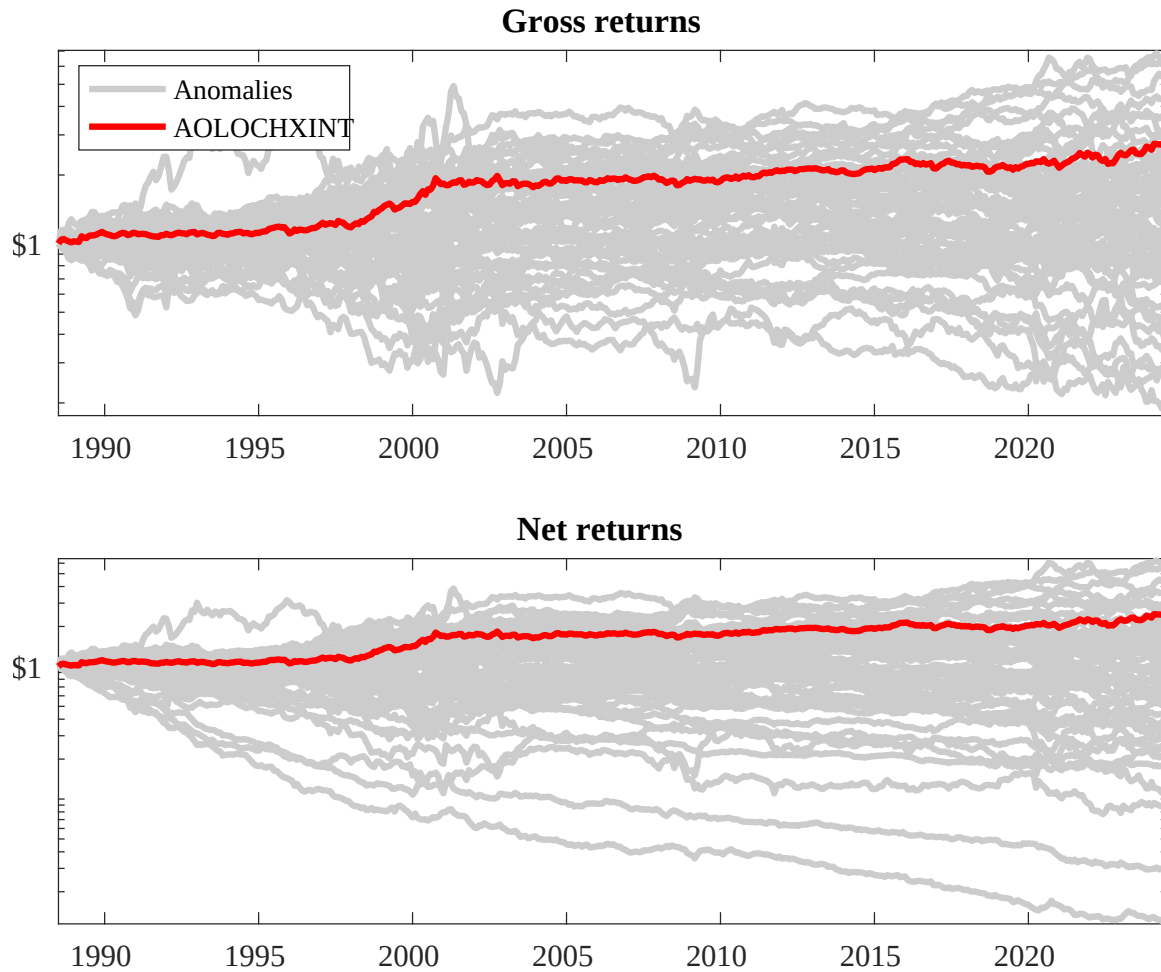
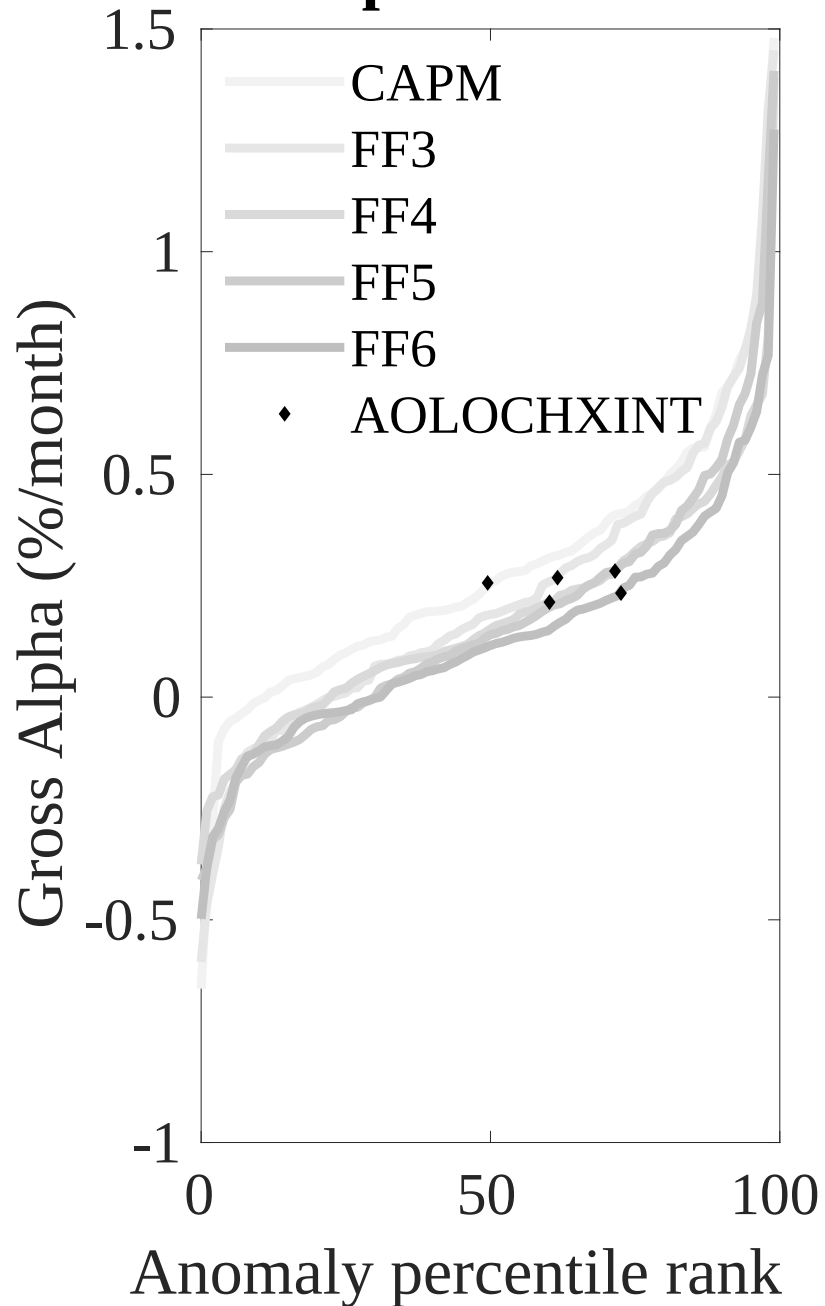


Figure 3: Dollar invested.

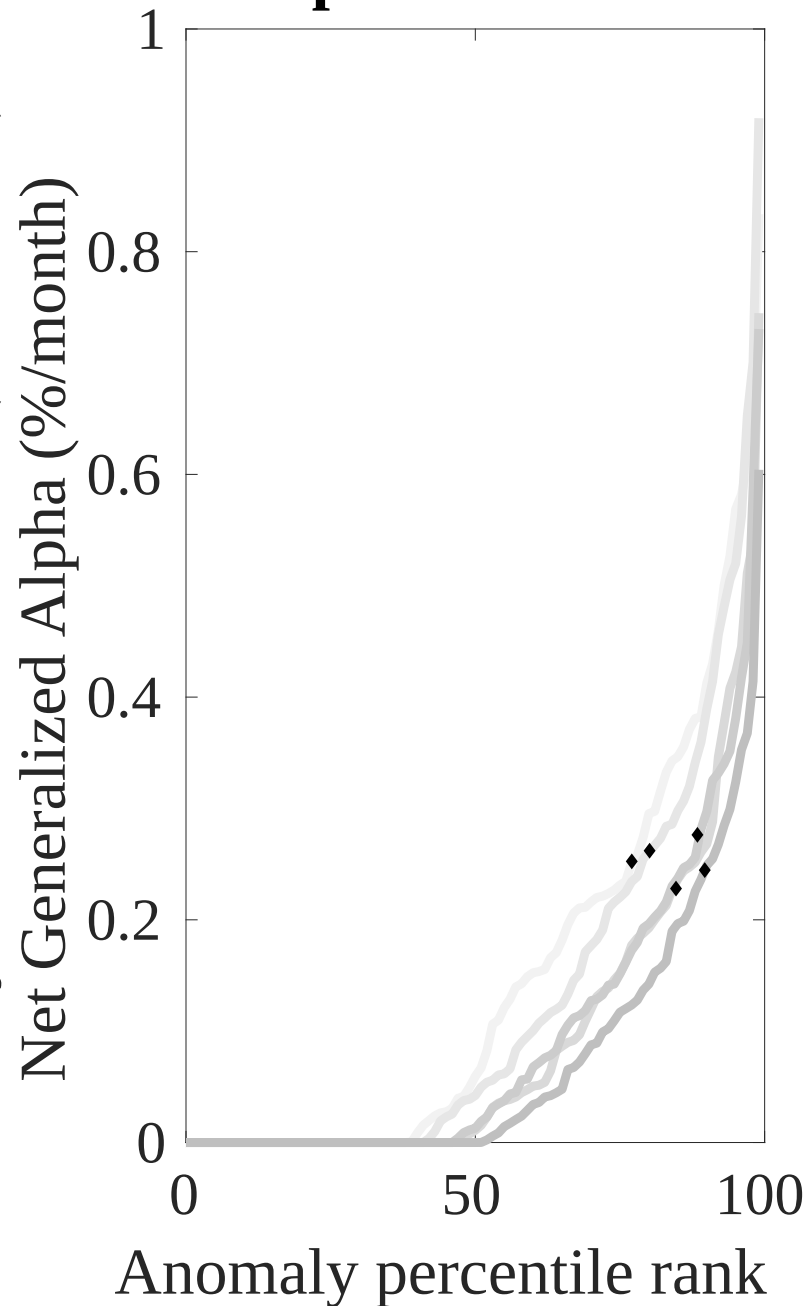
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



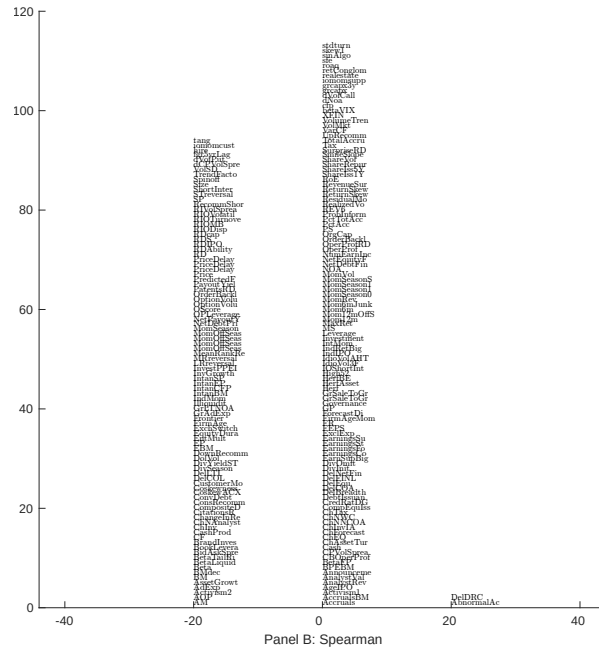
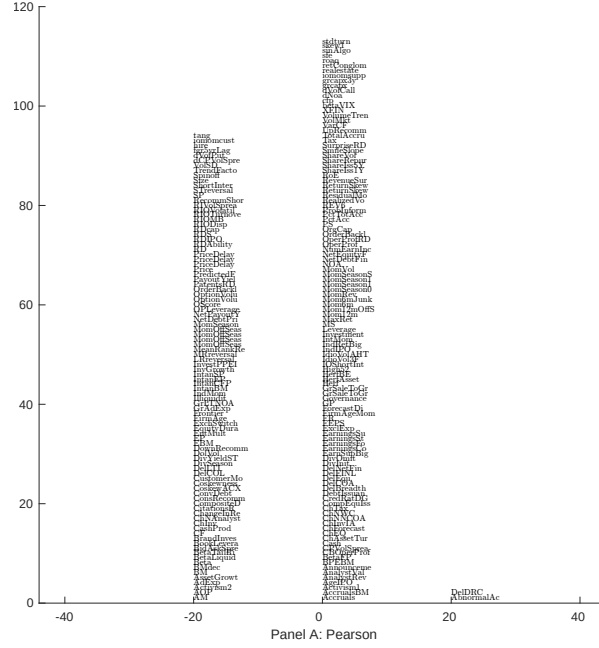


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

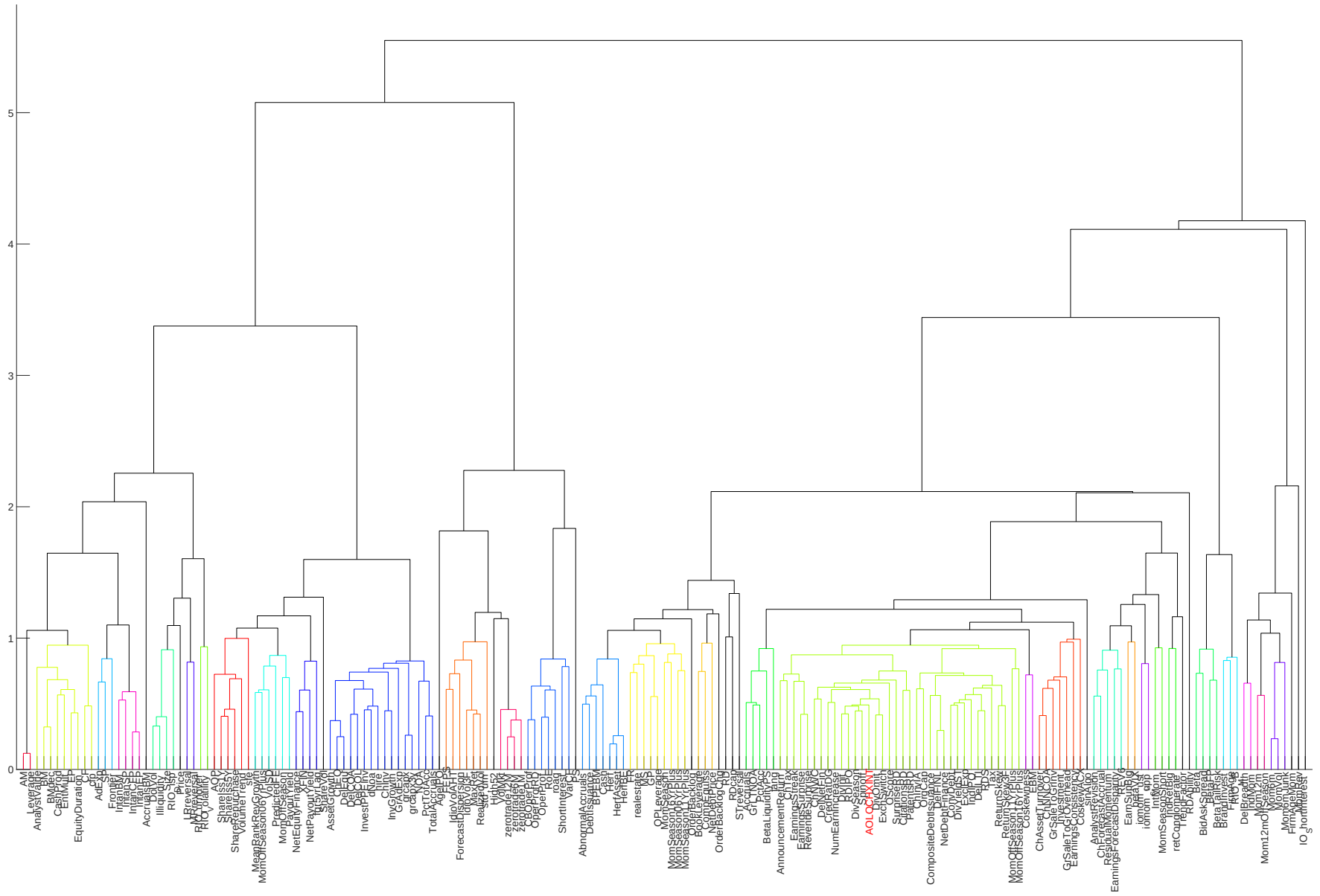


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

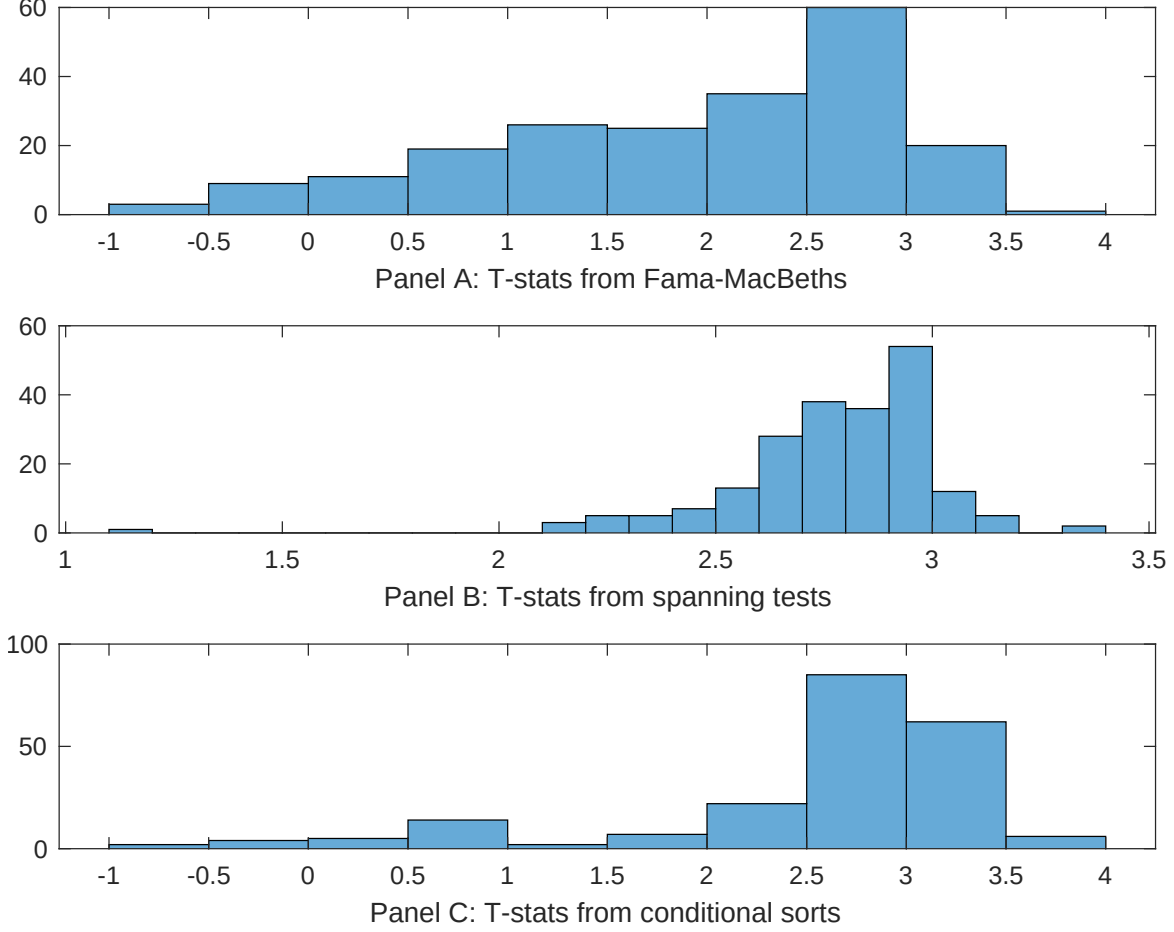


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DSF conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSF. Stocks are finally grouped into five DSF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSF trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSF. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSF}DSF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.95]	0.66 [1.85]	0.11 [3.87]	0.11 [3.90]	0.11 [2.97]	0.11 [3.86]	0.24 [3.38]
DSF	0.35 [3.05]	0.80 [1.59]	0.25 [1.68]	0.31 [2.71]	0.36 [3.21]	0.26 [2.21]	0.28 [0.42]
Anomaly 1	0.19 [4.46]						1.00 [0.58]
Anomaly 2		0.14 [4.58]					-0.25 [-3.26]
Anomaly 3			0.44 [4.46]				0.20 [6.41]
Anomaly 4				0.14 [5.38]			0.20 [2.25]
Anomaly 5					0.45 [0.88]		-0.75 [-0.81]
Anomaly 6						0.47 [1.64]	0.67 [1.05]
# months	432	415	427	432	432	432	267
$\bar{R}^2(\%)$	0	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DSF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating liabilities, Book-to-market and accruals, Efficient frontier index, Change in Taxes, Price, Total accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.54]	0.26 [2.99]	0.26 [2.93]	0.26 [3.05]	0.26 [2.96]	0.26 [3.03]	0.24 [2.71]
Anomaly 1	-12.60 [-3.09]						-8.98 [-2.02]
Anomaly 2		-1.21 [-0.96]					-1.07 [-0.85]
Anomaly 3			-0.40 [-0.15]				0.77 [0.27]
Anomaly 4				17.50 [5.03]			15.42 [4.24]
Anomaly 5					0.34 [0.10]		-0.42 [-0.12]
Anomaly 6						-7.57 [-1.87]	-1.45 [-0.33]
mkt	-1.62 [-0.75]	-1.95 [-0.89]	-2.03 [-0.92]	-3.61 [-1.68]	-1.94 [-0.86]	-1.74 [-0.80]	-3.17 [-1.41]
smb	-7.48 [-2.39]	-5.70 [-1.78]	-6.07 [-1.79]	-6.38 [-2.09]	-6.63 [-1.51]	-6.89 [-2.19]	-6.80 [-1.59]
hml	7.07 [1.65]	2.40 [0.58]	1.19 [0.30]	3.45 [0.92]	0.86 [0.22]	0.17 [0.05]	8.39 [1.76]
rmw	2.44 [0.62]	1.36 [0.34]	1.90 [0.48]	1.98 [0.51]	2.33 [0.52]	0.52 [0.13]	1.05 [0.23]
cma	-10.72 [-1.81]	-16.70 [-2.97]	-17.09 [-3.03]	-15.25 [-2.81]	-17.38 [-3.10]	-12.10 [-1.95]	-9.37 [-1.48]
umd	7.23 [3.74]	7.10 [3.40]	7.55 [2.86]	3.34 [1.60]	8.07 [2.73]	7.20 [3.66]	2.89 [0.91]
# months	432	428	428	432	432	432	428
$\bar{R}^2(\%)$	10	8	8	13	8	9	13

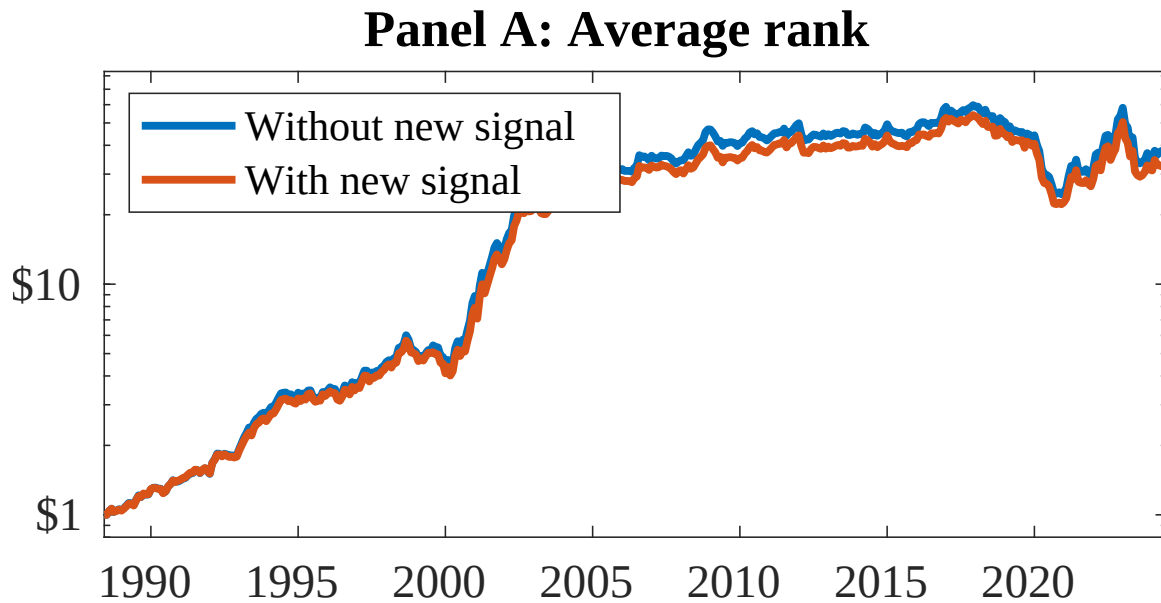


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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